

Proving Propositions 1 and 2 from Lecture on Feb. 17 (2015)

Proposition 2, stated in lecture today, was a restatement of Problem 17 from Section 1 (Page 180) of Hungerford:

Proposition 2.

- (a) If R is a ring with identity and A is a (left) R -module then there are submodules $A_1, A_2 \subseteq A$ with A_1 unitary, $RA_2 = \mathbf{0}$, and $A \cong A_1 \oplus A_2$.
- (b) If A' is a (left) R -module with $A' \cong A'_1 \oplus A'_2$, where A'_1 is unitary and $RA'_2 = \mathbf{0}$, and $f : A \rightarrow A'$ is an R -module homomorphism, then, using the notation of Part (a), $f(A_1) \subseteq A'_1 \oplus \mathbf{0}$ and $f(A_2) \subseteq \mathbf{0} \oplus A'_2$.
- (c) If the map f from Part (b) is an epimorphism then so are $f|_{A_1}$ and $f|_{A_2}$.

I correct the proof of Part (b) below and, for convenience, I include also the proofs of Parts (a) and (c). In correcting the proof of Part (b) I also took the opportunity to clarify the statement of Proposition 2 (and Exercise 17 of Hungerford): Since f maps to a direct sum, it makes more sense to say $f(A_1) \subseteq A'_1 \oplus \mathbf{0}$ than $f(A_1) \subseteq A'_1$ (although one could certainly *define* the latter to be short-hand for the former).

Proof of (a): Set $A_1 := \{1_R a \mid a \in A\}$ and $A_2 := \{a \in A \mid 1_R a = 0\}$. Clearly, A_1 is a submodule and, for any $b = 1_R a \in A_1$, we obtain $1_R b = 1_R 1_R a = 1_R a = b$. So A_1 is a unitary submodule.

Note also that, for any $b \in A_2$, and any $r \in R$ we have $rb = r1_R b = r\mathbf{0} = \mathbf{0}$. So $RA_2 = \mathbf{0}$.

Finally, since $a = 1_R a + (a - 1_R a)$, and $1_R a$ and $a - 1_R a$ clearly lie in A_1 and A_2 respectively, we obtain $A = A_1 + A_2$. So to finish, we need only check that $A_1 \cap A_2 = \{\mathbf{0}\}$. But this is easy to show since $b \in A_1 \implies b = 1_R a$. So assuming $b \in A_2$ in addition implies that $1_R 1_R a = \mathbf{0}$ and thus $1_R a = \mathbf{0}$, i.e., $b = \mathbf{0}$. ■

Proof of (b): If $a \in A_1$ then $1_R f(a) = f(1_R a) = f(a)$. So, writing $f(a) = (\alpha'_1, \alpha'_2)$ for some $\alpha'_1 \in A'_1$ and $\alpha'_2 \in A'_2$, we obtain $1_R f(a) = 1_R(\alpha'_1, \alpha'_2) = (1_R \alpha'_1, 1_R \alpha'_2) = (\alpha'_1, 0)$. So $f(a)$ indeed lies in $A'_1 \oplus \mathbf{0}$.

Now, if $a \in A_2$, then $1_R f(a) = f(1_R a) = f(\mathbf{0}) = (\mathbf{0}, \mathbf{0})$. So writing $f(a) = (\alpha'_1, \alpha'_2)$ tells us that $1_R(\alpha'_1, \alpha'_2) = (1_R \alpha'_1, 1_R \alpha'_2) = (\alpha'_1, 0)$. In other words, $\alpha'_1 = 0$ and thus $f(a) \in \mathbf{0} \oplus A'_2$. ■

Proof of (c): If f is an epimorphism then we can express any $a \in A'_1 \oplus \mathbf{0}$ in the form $f(a_1 + a_2) = a$ for some $a_1 \in A_1$ and $a_2 \in A_2$. But $f(a_1 + a_2) = f(a_1) + f(a_2)$. So, by Part (b), $f(a_2) \in \mathbf{0} \oplus A'_2$ and $f(a_1) \in A'_1 \oplus \mathbf{0}$, and thus $f(a_1) = a$. In other words, $f|_{A_1}$ is an epimorphism. By a completely symmetric argument we also obtain that $f|_{A_2}$ is an epimorphism. ■

Proposition 1, whose proof I omitted today, is stated and proved below.

Proposition 1. If R has an identity and P is a unitary (left) R -module then P is projective if and only if the following condition holds: For any unitary (left) R -modules A and B , and R -module homomorphisms $f : P \rightarrow B$ and $g : A \rightarrow B$ with the bottom row of the diagram below exact:

$$\begin{array}{ccc} & P & \\ \swarrow h_1 & \downarrow f_1 & \\ A_1 & \xrightarrow{g_1} B_1 & \longrightarrow \mathbf{0} \end{array} \quad (1)$$

there exists a homomorphism $h_1 : P \rightarrow A_1$ making the diagram (1) commute.

Proposition 1 is alluded to in the middle of Page 191 but I found the the argument for the assertion somewhat murky. So I've spelled it out a bit more in the proof below.

Proof of Proposition 1: ($\Rightarrow$): The definition of projective module immediately implies the stated condition (which is just a special case of the original definition). ■

($\Leftarrow$): Now we show that our stated restricted definition implies the full (original) definition of projectivity. In particular, suppose A and B are (left) R -modules, and $f : P \longrightarrow B$ and $g : A \longrightarrow B$ are R -module homomorphisms such that the bottom row of the diagram below is exact:

$$\begin{array}{ccccc} & & P & & \\ & & \downarrow f & & \\ A & \xrightarrow{g} & B & \longrightarrow & \mathbf{0} \end{array}$$

We need to prove that there is an R -module homomorphism $h : P \longrightarrow A$ with $f = g \circ h$. We will use the unitariness of P to show that such an h can be built out of a simpler map h_1 that is already guaranteed exist.

In particular, Part (a) of Proposition 2 tells us that there are (left) sub-modules $A_1, A_2 \subseteq A$ and $B_1, B_2 \subseteq B$ with $A \cong A_1 \oplus A_2$, $B \cong B_1 \oplus B_2$, A_1 and B_1 both unitary, and $RA_2 = \mathbf{0} = RB_2$. Part (b) also tells us that $f(P) \subseteq B_1$, $g(A_1) \subseteq B_1$, and Part (c) tells us that $g_1 := g|_{A_1}$ is an epimorphism since g is an epimorphism. Since we are assuming the universality property (1), there must then be a morphism $h_1 : P \longrightarrow A_1$ with $f(x) = g_1(h_1(x))$ for all $x \in P$.

Taking $h := h_1$ we then clearly obtain $g(h(x)) = g_1(h_1(x)) = f(x)$ for all $x \in P$, so we are done. ■